Reference B of CHAPTER 1

The Efficient Market Hypothesis

From Chapter 11 of 'Investments' by Zvi Bodie, Alex Kane, and Alan J. Marcus, 2023, Mc Graw Hill

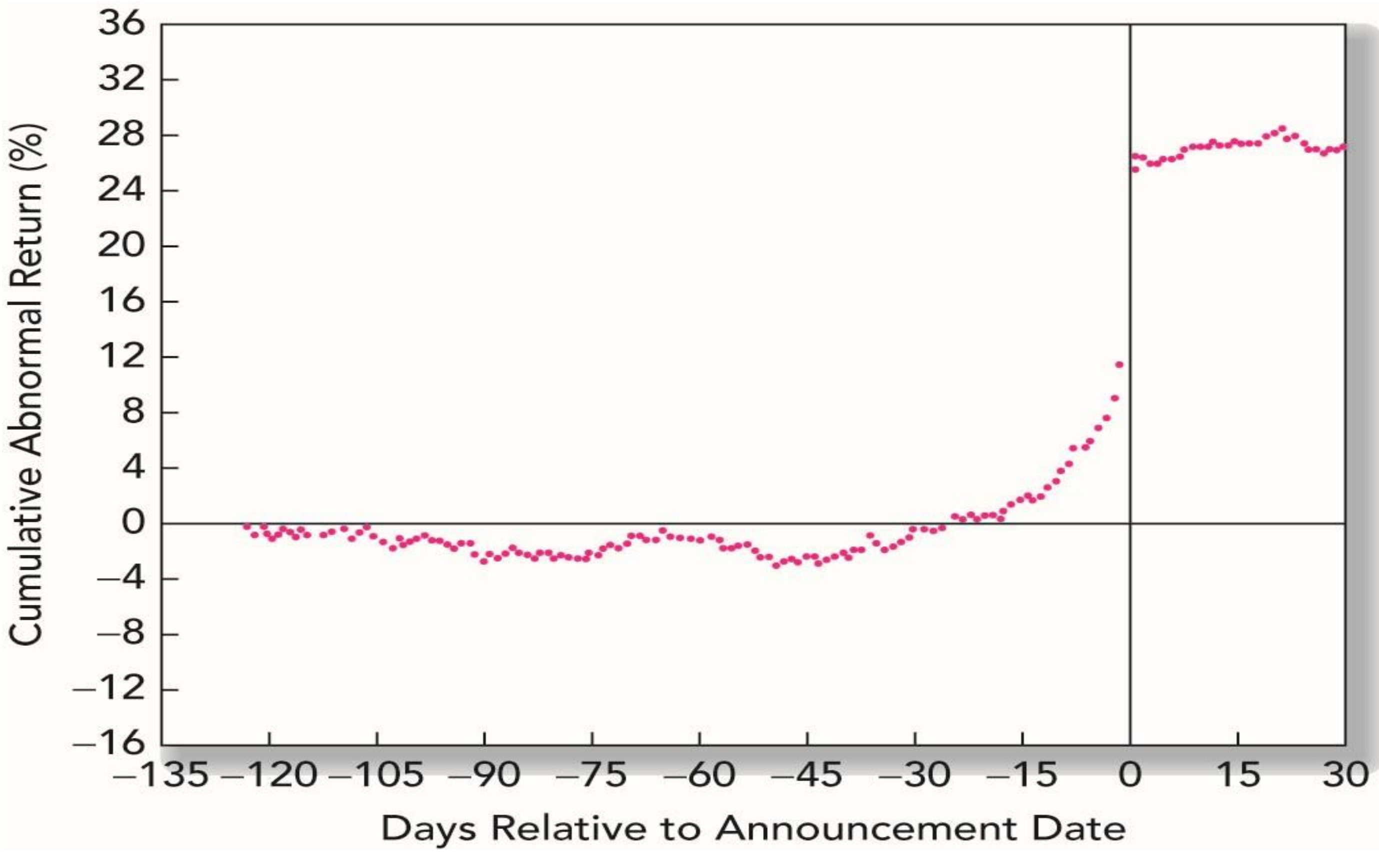
Random Walks and the Efficient Market Hypothesis (1)

- M. Kendall (1953) found that he could identify no predictable patterns in stock prices. Prices evolve randomly.

- Stock prices should follow a *random walk*. → It should be *random* and *unpredictable*.
  - Do security prices reflect information ?

→ Efficient Market Hypothesis (EMH)

Figure. Cumulative Abnormal Returns before Takeover Attempts: Target Companies



There is no further drift in prices *after* the announcement date.

Random Walks and the Efficient Market Hypothesis (2)

- Competition as the source of efficiency
  - Stock prices fully and accurately reflect all publicly available information in a competitive market.
  - Once information becomes available, market participants analyze it.
  - Competition assures prices reflect information.

Random Walks and the Efficient Market Hypothesis (3)

- Versions of the EMH
  - **Weak-form**
    - Stock prices already reflect all information that can be derived by examining *market trading data* (e.g. past prices, trading volume, short interest)
  - **Semistrong-form**
    - *All publicly available information* (e.g. past prices, data on firm's product line, quality of management, B/S composition, patents held, earning forecast, accounting practices) regarding the prospects of a firm must be reflected already in the stock prices.

Continued…

- **Strong-form**
  - Stock prices reflect *all information relevant to the firm*, even including information available only to company insiders.
  - Rule 10b-5 of the Security Exchange Act (1934) sets limit on trading by corporate officers, directors, and substantial owners, requiring them to report trades to the SEC.
- The hierarchy of the information sets

strong-form EMH (?) semistrong-form EMH
(?) weak-form EMH

Implications of the EMH (1)

- Technical Analysis
  - Is essentially the search for recurrent and predictable patterns in stock prices.
  - *Chartists*: study records or charts of past stock prices, hoping to find patterns to make a profit

    → resistance levels (or support levels)
  - Weak-form EMH & technical analysis
- Fundamental Analysis
  - Using economic and accounting information to predict stock prices. (study of past earnings, examination of company B/S)
  - Semistrong-form EMH & fundamental analysis

Implications of the EMH (2)

- Active VS passive portfolio management
  - Active management: security analysis etc.
  - Passive Management: buy and hold index funds and ETFs etc.
  - Proponents of the EMH believe that active portfolio management is largely wasted effort.
  - *Hybrid strategy*

    = Passive core (indexed position)

    + Actively managed portfolio

Are Markets Efficient? (1)

- The EMH has never been widely accepted on Wall Street.
  - Can security analysis improve investment performance?
  - Three issues: magnitude issue, selection bias issue, lucky event issue.
- The magnitude issue: measure of performance
  - Are markets efficient? → *How* efficient are the market?
  - Example: a portfolio manager overseeing $5 bil., who can improve performance by only 0.1% per year. Can we statistically measure her contribution?

  (Probably not. A 0.1% contribution would be swamped by the yearly volatility of the market. A small increase in performance would be hard to detect.)

Are Markets Efficient? (2)

- The selection bias issue
  - The outcomes we are able to observe have been preselected in favor of failed attempt (i.e., investment scheme not giving abnormal returns).

  → We cannot fairly evaluate the true ability of portfolio managers to generate winning stock market strategies. (opponents)
- The lucky event issue
  - An investor with a fantastic investment performance may *not* be a virtual evidence disproving the EMH. (it may be just lucky.)
  - We should check whether such successful investors can repeat their performance.

Are Markets Efficient? (3)

- Weak-form tests: patterns in stock returns
  - Returns over short horizons
    - Serial correlation of stock market returns (Conrad and Kaul 1988, Lo and Mackinlay 1988): Tendency for stock returns to be related to past returns. → cannot clearly reject the EMH.(But weekly returns of the NYSE stocks had *positive* serial correlation.)
    - Momentum effect (Jegadeesh and Titman 1993): good or bad recent performance of particular stocks continues over time.

Continued…

- Returns over long horizons
  - Tests of long horizons have found negative long term serial correlation. → call it ‘fads’ hypothesis → Stock market might overreact to relevant news.
  - Reversal effect (DeBondt and Thaler 1985, Chopra et al. 1992): The best (worst)-performed stocks seem to underperform (outperform) in following multiyear period.
  - *Short-term overreaction*, which causes momentum in prices, may lead to *long-term reversals*, when market recognizes its past error.

Are Markets Efficient? (4)

- Semistrong Tests: Market Anomalies
  - The price-earnings (P/E) effect (Basu 1977): Portfolios of low PER stocks have higher returns than do high PE portfolios. The P/E Effect holds up even if returns are adjusted for portfolio beta.

  - The small-firm effect (or size effect, Banz 1981)
    - The neglected-firm effect (Arbel and Strebel 1983): Small firms are neglected by large institutional traders. Information about smaller firms is less available.
    - The liquidity effects (Amihud and Mendelson 1986): Investors demand a rate-of-return premium to invest in less-liquid stocks.

Figure. Returns in Excess of Risk-Free Rate and in excess of the Security Market Line for 10 Size-Based Portfolios (1926 – 2008)

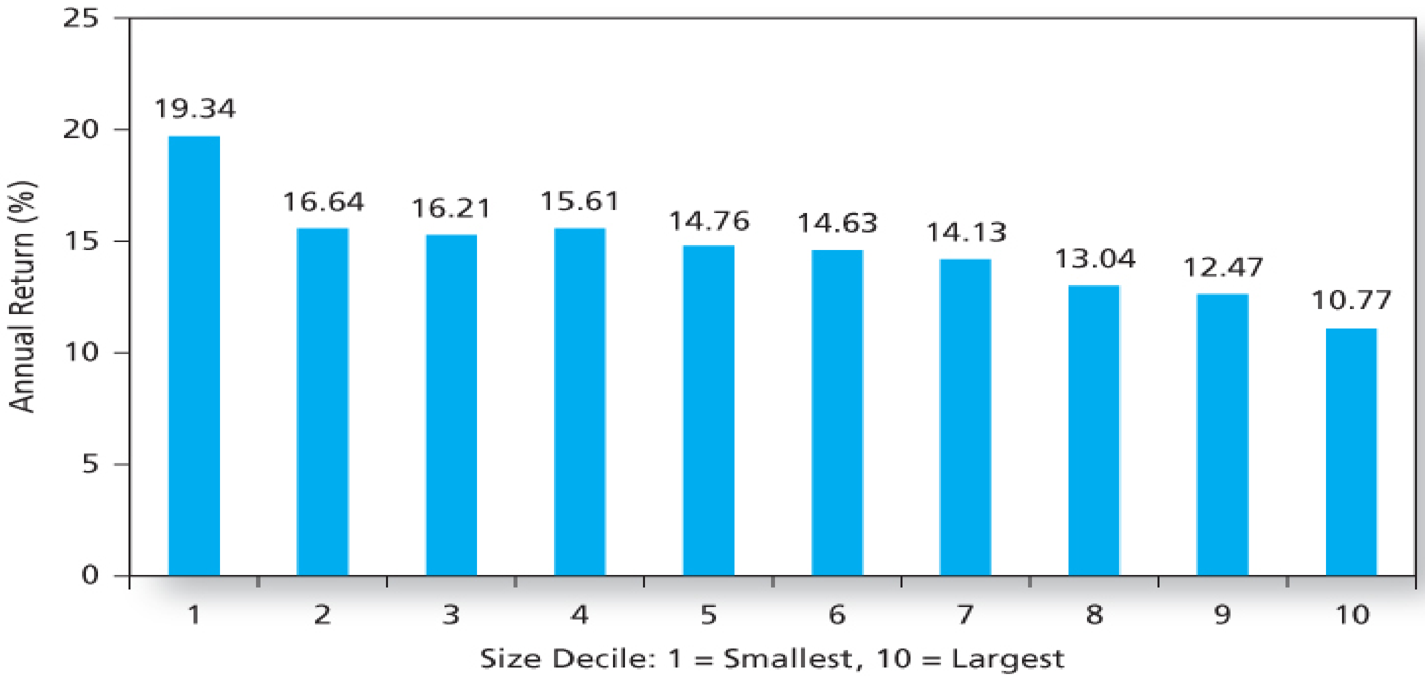


Figure 11.3 Average annual return for 10 size-based portfolios, 1926–2008

Source: Authors' calculations, using data obtained from Professor Ken French's data library at **http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.**

Continued…

- Book-to-market effects (Fama and French 1992)
  - Higher book-to-market ratio implies higher average monthly return.
  - They also found that after controlling for the size and book-to-market effects, beta seemed to have *no* power to explain average security returns.

- Post-earnings-announcement price drift (Ball and Brown 1968)
  - They found the apparently sluggish response of stock prices to firm's earnings announcements, different from the property of efficient Markets.

Figure. Average Monthly Returns as a Function of the Book-To-Market Ratios (1926 – 2008)

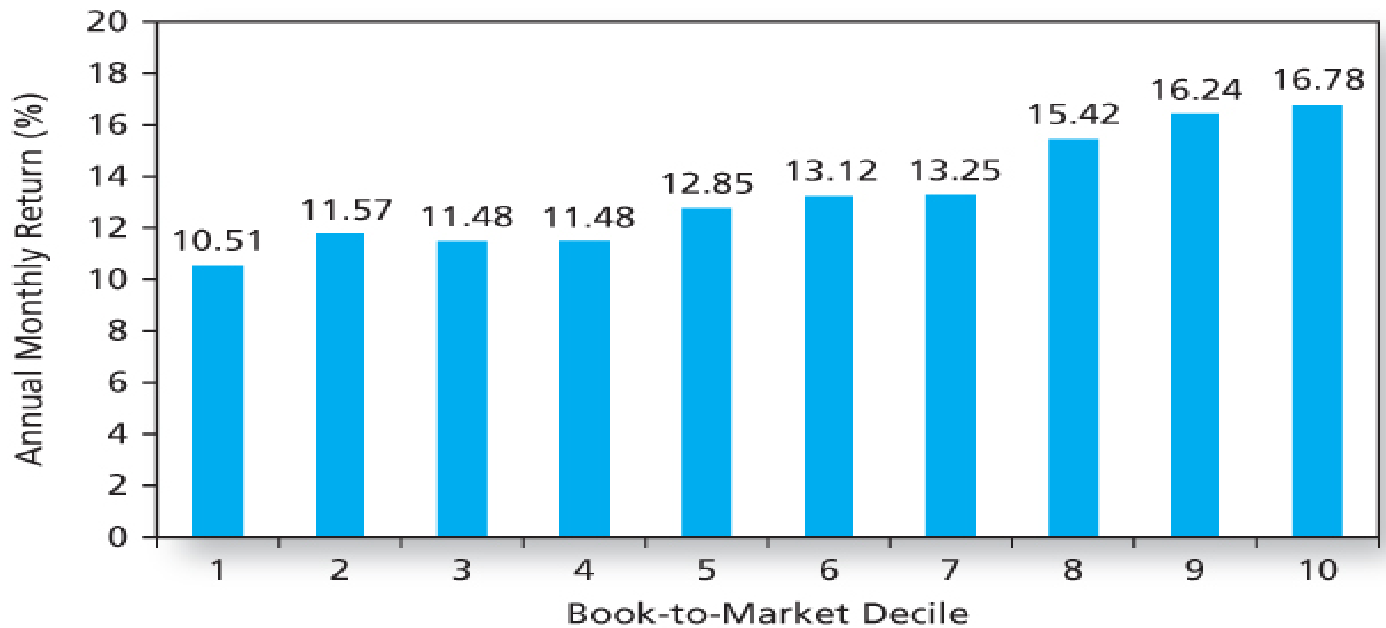
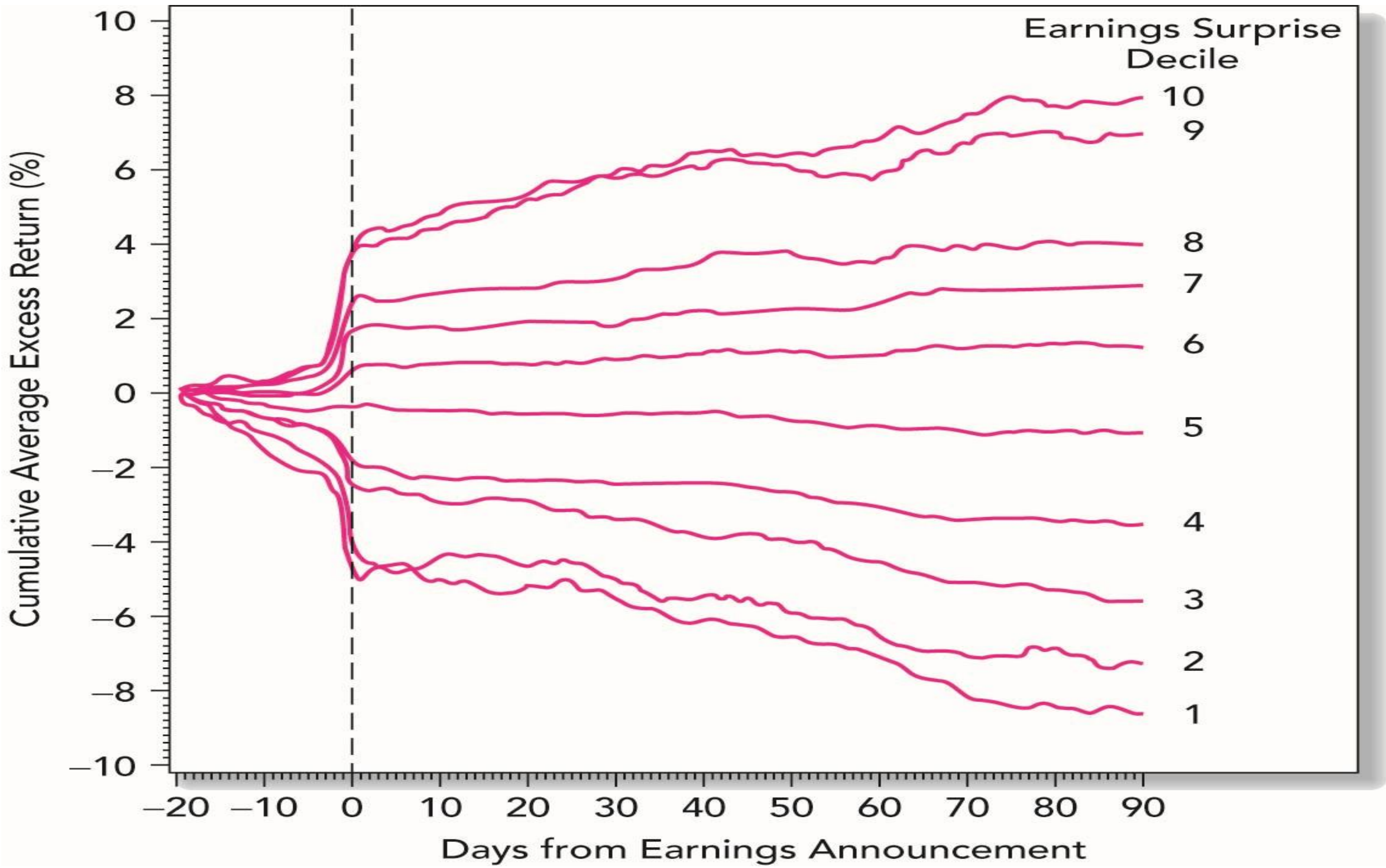


Figure 11.4 Average return as a function of Book-to-market ratio, 1926–2008

Source: Authors' calculations, using data obtained from Professor Ken French's data library at **http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.**

Figure. Cumulative Abnormal Returns in Response to Earnings Announcements



Are Markets Efficient? (5)

- Strong-form tests: Inside information

  - We do not expect markets to be storing-form efficient.

  → Regulation for insider trading (SEC)

  - Can other investors benefit by following insiders' trade?

  → Seyhun (1986) found that following insider transactions would be to no avail. (Abnormal return are not sufficient)

Figure. Persistence of Mutual Fund Performance

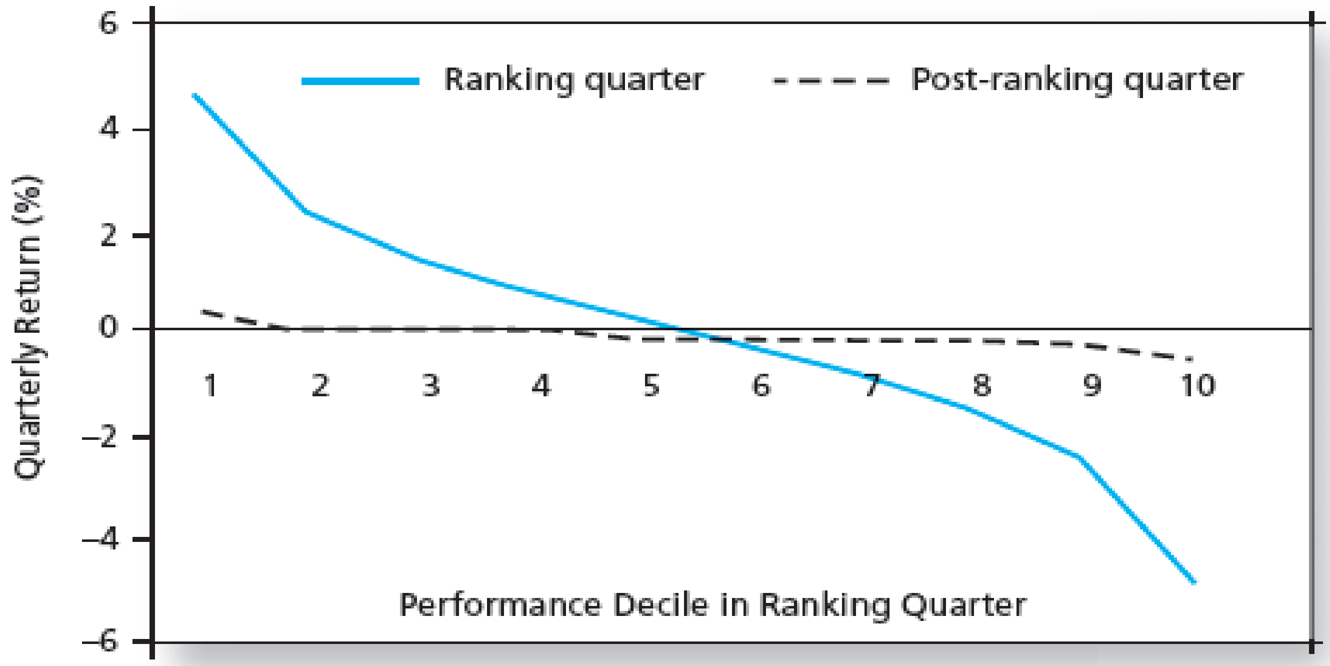
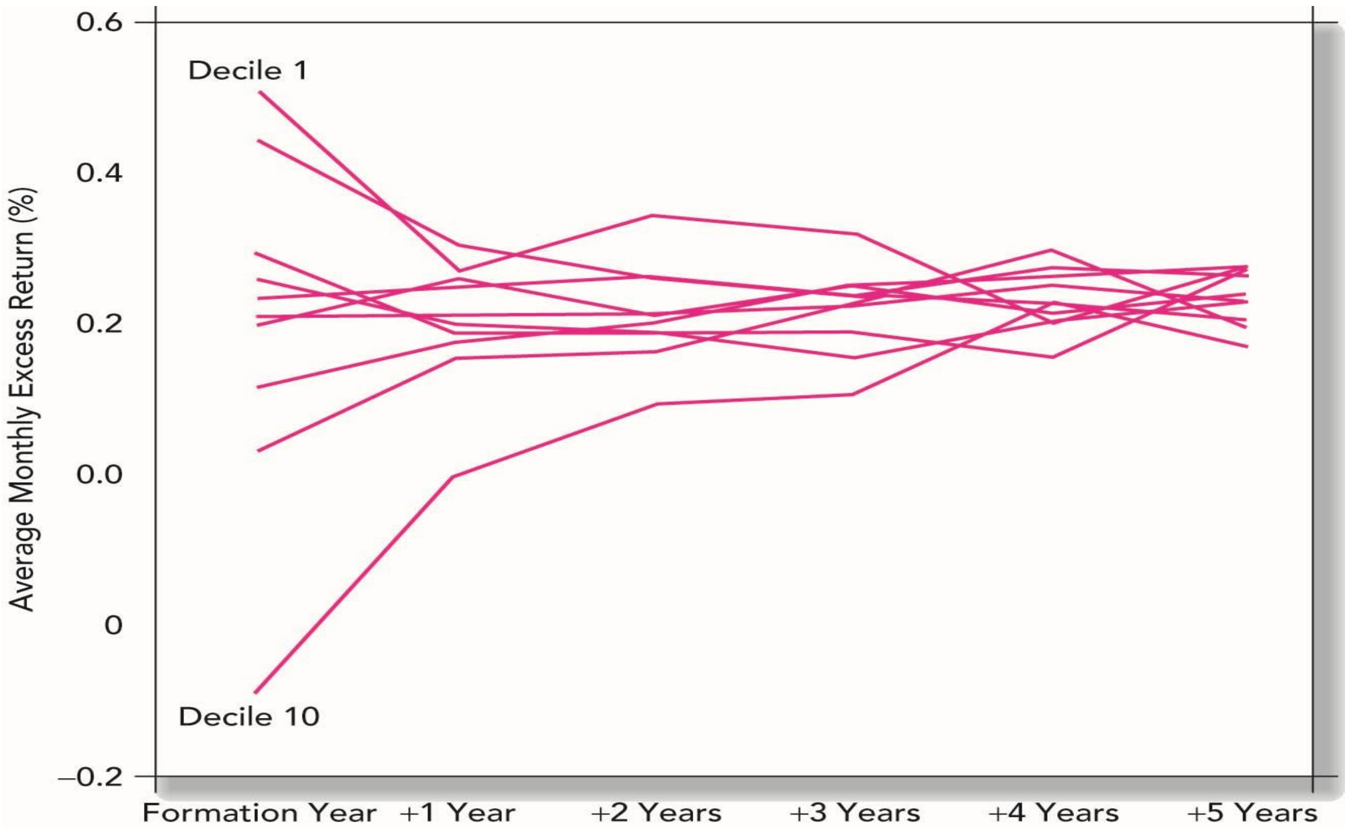


Figure 11.8 Risk-adjusted performance in ranking quarter and following quarter

Even if managers are skilled, alphas will be short-lived.

Figure. Persistence of Mutual Fund Performance



Carhart (1997) found that performance in future periods is almost independent of earlier-year returns.

So, Are Markets Efficient?

- The performance of professional managers is broadly consistent with market efficiency.
- Most managers do not do better than the passive strategy.
- There are, however, some notable superstars:
  - Peter Lynch, Warren Buffett, John Templeton, George Soros